

Chap 2

Date

No.

1 byte = 8 bits sizeof $\Rightarrow$ bytes

Subtraction:

$$\begin{array}{r} 22 \\ -11 \\ \hline 100 \\ -11 \\ \hline 001 \end{array}$$

$$\begin{array}{r} 2 \\ -112 \\ \hline 100(2) \\ -11(2) \\ \hline 001(2) \end{array}$$

if there's n bit. for 2's = $-2^{n-1} \sim 2^{n-1}-1$

Decimal Value of a 2's complement notations

In general. for a n-bit integer,

$$N = [\text{num represented by bits } 0 \text{ to } n-2] - [\text{bit}(n-1) \times 2^{n-1}]$$

Unsigned Integer

$0 \sim 2^n - 1$ (the number is called an unsigned integer because the MSB is no longer used as a sign bit. Now the MSB carries a weight for unsigned integer) By default, integers are signed

Overflow

XOR (exclusive or " $\wedge$ ": same = 0, different = 1)

★★★ $C_{n-1} \wedge C_{n-2} = 1 \Rightarrow$ wrong result 进位不一样就是错误的结果

① carry bit (进位位)

指在加法中,从MSB产生的进位

在无符号数的计算中,若 carry bit = 1 $\Rightarrow$ 错误; 0 $\Rightarrow$ 正确

② overflow bit (溢出位)

是针对有符号数(补码表示)的. 在补码运算中,是否溢出不能只看 carry

$$\text{overflow} = C_{n-1} \wedge C_{n-2}$$

eg. 8 bit $\begin{array}{c} \text{C}_{n-1} \\ \text{C}_7 \end{array} \begin{array}{c} \text{C}_6 \\ \text{C}_6 \end{array}$

C_{n-2} : 进入符号位的进位

C_{n-1} : 从符号位产生的进位

288.16

Convert from Binary to Octal: group of 3 bits

$$0110101012 = 3258$$

0... 11进制 $012 = 1 \times 8^1 + 2 \times 8^0 = 10$
0X/0x... 十六进制 $0X1A = 1 \times 16^1 + 10 \times 16^0 = 26$

2 $\rightarrow$ 8/16 3位-组/4位-组

都是从小数点开始向左向右分组; 若不够的话最左和最右补0

Base 10 $\rightarrow$ Base R

$$106_{10} \rightarrow 6A_{16}$$

$$16 \overline{) 106}$$

$$16 \overline{) 6} \dots 10 \uparrow \text{LSB}$$

$$0 \dots 6 \downarrow \text{MSB}$$

A'ZONE

		32	
	'\0'	0	
ASCII	'\n'	10	
	'0'~'9'	$48_{10} \sim 57_{10}$ +10=-1	$30_{16} \sim 39_{16}$
	Uppercase	$65_{10} \sim 90_{10}$ +26=-1	$41_{16} \sim 5A_{16}$
	Lowercase	$97_{10} \sim 122_{10}$ +26=-1	$61_{16} \sim 7A_{16}$
	Low $\rightarrow$ Upper: $-32_{10} / -20_{16}$		

7 bit code for 128 characters, the 8th bit (bit 7) is used for error checking during data transmission.

只有128个字符,但自身范围还是 $-2^7 \sim 2^7-1$,即 $-128 \sim 127$,只是负数无字符意义